1 The kinetic energy E of an object rotating about its centre of mass is calculated using

$$E = \frac{1}{2} k\omega^2$$

where ω is the angular speed of rotation, and k is a constant.

What is the unit for k?

A kg m^2 **B** kg m s^{-1} **C** kg m s^{-2} **D** kg m^2 s^{-4}

Ans: **A**

$k = \frac{2E}{\omega^2}$ → [k] = (kg m^2 s^{-2}) (rad s^{-1})$^{-2}$ = kg m^2

2 The following measurements were taken of a uniform metal disc.